

```
$ mysql -e "SHOW CREATE TABLE orders" --normalize
```



// 1NF → 2NF → 3NF □□□□□

1NF

□□□

2NF

□□□□□□

3NF

□□□□□□□

□□□□

OLAP □□□

\$ cat schema-plan.md



01 / 

□□□□□□□□

02 / 

1NF    → 2NF → 3NF

03 / 

JOIN □□□

04 / 

OLTP vs OLAP □□□

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PART 01



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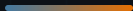
order_id	customer_name	customer_email	product_name	price
1	□□	<u>zhang@example.com</u>	□□□□□	30000
2	□□	<u>zhang@example.com</u>	□□	500
3	□□	<u>li@example.com</u>	□□	1200

□□□□

□□□□□□□□ email □□□□

□□□□

□□□ email□□ UPDATE □□□□



ROOT CAUSE



CHECKPOINT_01

データベースの UPDATE 37 回実行後 3 分経過

A データが更新された

B データが削除された

C データが追加された

PART 02



1NF → 2NF → 3NF → 4NF → 5NF

1NF

NO DUPLICATES

BEFORE —

student	courses
	Maths, Maths, Maths, Maths, Maths, Maths, Maths, Maths

AFTER — 1NF

student	course
	Maths
	Maths
	Maths

2NF **NOGRAPH** ☐ ☐ ☐ ☐ ☐ ☐

BEFORE — □□□□

order_id	product_id	product_name	qty
1	P1	□□	1
2	P1	□□	3

AFTER — □□□□□

order_id	product_id	qty
1	P1	1
product_id	product_name	
P1	□□	

3NF

NO SLIP

□□□□□□

student_id → dept_id → dept_name — dept_name — dept_id —

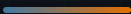
BEFORE — □□□□

student_id	name	dept_id	dept_name
S1	□□	D1	□□□
S2	□□	D1	□□□

AFTER — □□□□□

student_id	name	dept_id
S1	□□	D1
dept_id	dept_name	
D1	□□□	

□□□□□□□□□□ — □□□□□□□□□□



00000000000000000000 3NF 0000 0000000000000000

PART 03



□□□□□□JOIN □□□

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JOIN □□
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JOIN □□□□□

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OLTP vs OLAP



OLTP	OLAP	OLTP
OLTP	OLAP	OLTP
OLTP	OLAP	OLTP
OLTP	OLAP	OLTP
OLTP	OLAP	OLTP
OLTP	OLAP	OLTP

STRATEGY

OLTP is designed for high transaction volume and fast response time. OLAP is designed for complex queries and data analysis.



→



1NF     → 2NF     → 3NF    

3

NF

1

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OLTP □□□□OLAP □□□□□□□□

Schema normalized.

XXXXXXXXXXXX JOIN XXXXX

```
$ mysql -e "EXPLAIN SELECT ..."
```

1NF → 2NF → 3NF → XXXXXXXX